Abelian Extensions Problem Set 2

1. Suppose p is an odd prime and let ζ_p be a primitive p-th root of unity.. Show that

$$g = \sum_{k=1}^{p-1} \left(\frac{k}{p}\right) \zeta_p^k$$

satisfies $g^2 = \left(\frac{-1}{p}\right)\sqrt{p}$. (Here, $\left(\frac{k}{p}\right)$ denotes the Legendre symbol. You may use the fact that $\zeta_p^{p-1} + \zeta_p^{p-2} + \cdots + \zeta_p + 1 = 0$.)

2. Use the previous exercise to show that any quadratic extension $L/\mathbb{Q}$ is contained in a cyclotomic field. (You may use the fact that any quadratic extension is of the form $L = \mathbb{Q}\left(\sqrt{a/b}\,\right), a/b \in \mathbb{Q}$.)

3. Prove the first reduction step stated in today's lecture. Suppose that $L/\mathbb{Q}$ is a finite abelian extension.
   (a) Write $\mathrm{Gal}(L/\mathbb{Q})$ as a product of groups $\prod_{i=1}^{r} G_i$, where each G_i is an abelian group whose order is a prime power.
   (b) Using the product given above, find subgroups $H_j \subseteq \mathrm{Gal}(L/\mathbb{Q})$ each of whose *index* is a prime power.
   (c) Use the Galois correspondence to find the prime power extensions $L_1, L_2, \cdots, L_r$ whose compositum is L.
   (d) Show that if for each i, $L_i \subseteq \mathbb{Q}(\zeta_{n_i})$, then $L \subseteq \mathbb{Q}(\zeta_N)$ for some N.

4. If $\beta \in \mathbb{C}$ is an algebraic number (i.e., is algebraic over $\mathbb{Q}$), with $\deg(m_{\beta,\mathbb{Q}}) = n$, then $\mathbb{Z}[\beta] := \{a_0 + a_1\beta^1 + \cdots + a_{n-1}\beta^{n-1} | a_i \in \mathbb{Z}\}$. Go to LMFDB, and click on the "Number Fields" link in the menu. Look through the terms on that page and find the name for a ring of integers of the form $\mathcal{O}_L = \mathbb{Z}[\beta]$. Then, give an example of a field $L/\mathbb{Q}$ for which $\mathcal{O}_L \neq \mathbb{Z}[\beta]$ (Note: You're not being asked to find such a field on your own - instead, you're being asked to spend some time looking around LMFDB.)

5. A subset $A \subseteq \mathcal{O}_L$ is an ideal of $\mathcal{O}_L$ if A is a subgroup of $\mathcal{O}_L$ (under addition) and for any $a \in A$ and $b \in \mathcal{O}_L$, $ab \in \mathcal{O}_L$. If you have not worked with ideals before (or if you need a refresher), prove the following:
   (a) If $n \in \mathbb{Z}$, then $n\mathbb{Z} = \{nk : k \in \mathbb{Z}\}$ is an ideal of $\mathbb{Z}$.
   (b) For any $\mathcal{O}_L$, $\mathcal{O}_L$ is an ideal (referred to as the unit ideal). Show also that if $a \in \mathcal{O}_L$, then the set $(a) = \{ab | b \in \mathcal{O}_L\}$ is an ideal (this is called a principal ideal).

(c) If A and B are ideals of $\mathcal{O}_L$, then the ideal product AB defined by

$$AB = \left\{ \sum_{i=1}^{n} a_i b_i |\ a_i \in A \text{ and } b_i \in B \right\}$$

is an ideal of $\mathcal{O}_L$.

(d) If $A \subseteq \mathcal{O}_L$ is an ideal of $\mathcal{O}_L$, and if $\sigma \in \mathrm{Aut}(L/\mathbb{Q})$, then $\sigma(A) = \{\sigma(a) | a \in A\}$ is an ideal of $\mathcal{O}_L$. (Note: You should first show that σ maps $\mathcal{O}_L$ to $\mathcal{O}_L$.)

(e) If A, B are ideals of $\mathcal{O}_L$, then $\sigma(AB) = \sigma(A)\sigma(B)$. Note that this implies $\sigma(A^n) = \sigma(A)^n$ for all $n \in \mathbb{Z}+$. Conclude that if $p \in \mathbb{Z}$ is a prime, then (assuming the result that factorization of nonzero ideals in $\mathcal{O}_L$ is unique) if

$$p\mathcal{O}_L = \mathfrak{p}_1^{e_1} \mathfrak{p}_2^{e_2} \cdots \mathfrak{p}_g^{e_g}$$

is the factorization of $p\mathcal{O}_L$, then $\sigma\left(\mathfrak{p}_i^{e_i}\right) = \mathfrak{p}_j^{e_j}$ for some $\mathfrak{p}_j^{e_j}$ appearing in the factorization of $p\mathcal{O}_L$ (and that in this case, $e_i = e_j$.